

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. EC (CSE) Examination

May 2018

EC745/ EC711.02/ EC711.01 Principles of Wireless Networks

Date: 08.05.2018, Tuesday

Time: 1:30 p.m. to 4:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a)** What do you mean by Ad hoc Network? Give its applications. **[04]**
- (b)** Give the challenges of PRNETs. **[03]**
- Q - 2 Answer the following (Any Two)** **[14]**
- (a)** Enlist technical challenges of Ad hoc Network. Describe any three challenges in Ad Hoc Networks. **[07]**
- (b)** Describe neighbor table, tier table and device table for route calculation in case of packet radio network. **[07]**
- (c)** What is impact of mobility in packet radio network? Illustrate traffic profiles of Ad hoc network scenario. **[07]**
- Q - 3 (a)** Describe MACA- BI wireless media access protocol. How is it better than MACA protocol? **[07]**
- (b)** Explain the operation of media access with reduced handshake (MARCH) protocol with figure. **[07]**

OR

- Q - 3 (a)** Explain Hidden terminal and Exposed node problem. Briefly explain solution for exposed node problem. **[07]**
- (b)** Explain Cluster Switch Gateway Routing (CSGR) with necessary figure. **[07]**

SECTION – II

- Q- 4 (a)** Describe TCP Snoop. [03]
- (b)** Compare the sender initiative access protocol and receiver initiative access protocol. [02]
- (c)** What is the difference between Bluetooth technology and Infrared Technology? [02]
- Q - 5 (a)** Explain the Bluetooth protocol architecture with necessary figure. [07]
- (b)** Draw the Bluetooth piconet architecture and the Bluetooth Scattered architecture. [07]

OR

- Q - 5 (a)** Explain Service Coordinator based architecture with necessary figures. [07]
- (b)** Enlist Ad hoc Nomadic mobile applications. Describe any two in brief. [07]
- Q - 6 (a)** Describe Zone Routing Protocol (ZRP) with neat diagram. [06]
- (b)** Explain Service Location Protocol architecture with necessary figure. [06]
- (c)** Give the difference between 3G and 4G technologies. [02]

OR

- Q - 6 (a)** Describe Long Term Evolution (LTE) of 4G mobile technology. Discuss Voice over LTE (VOLTE). [06]
- (b)** Describe Wireless Routing Protocol (WRP) with figure. [06]
- (c)** Draw Internet Protocol Architecture. [02]
